

Supplementary Material: Physical-Reward Policy Optimization for Accelerator RTL

Mohamed Amine Hallam, Kuo-Kun Tseng, and Wenjie Pei

Abstract—This companion document preserves completed supporting studies, physical validation details, and reward diagnostics for the main paper. A worked RTL example and runnable package accompany a completed RF versus SFT comparison. Historical and post-primary results retain their original populations and measurement contracts; they are not pooled with the sealed primary estimate.

SI. WORKED RTL EXAMPLE AND MATCHED RF VERSUS SFT EVALUATION

A. What the example demonstrates

The accompanying runnable package connects the paper’s training objective to an inspectable hardware example and then to the complete policy comparison. The example uses `fir13_v6_8b`, the FIR design also discussed in the main paper, with fresh outputs from the completed post-primary campaign. For each policy, we select the most frequent functionally passing exact RTL string among its 48 draws on this design, breaking ties by the content hash. This selection is made after the experiment to explain the implementation structure; timing is not used to choose either string. The pair illustrates what a physical reward can favor, while the all-design analysis below measures whether the resulting distribution is better under the stated endpoint.

In the SFT implementation, registers retain delayed inputs and a combinational expression forms their weighted sum before the output register. In the RF implementation, each stage stores a partial weighted sum, giving a transposed FIR structure (Fig. S1). The selected strings occur in 17 SFT draws and 24 RF draws. Their measured bounded setup-period frequencies are 52.87 and 230.33 MHz, respectively. Both pass the same simulation gate, including repeated reset episodes and latency alignment. This concrete difference is consistent with the intended reward, but does not establish that the RF policy discovered a circuit absent from SFT support. The measurements belong to the new campaign and do not replace the earlier FIR measurements reported in the main paper.

B. How to inspect and run the package

After installing the dependency listed in `requirements.txt`, running `python run_demo.py` verifies the packaged hashes and recomputes the comparison from all archived draws. The output includes the policy means,

Mohamed Amine Hallam, Kuo-Kun Tseng, and Wenjie Pei are with the School of Computer Science and Technology, Harbin Institute of Technology (e-mail: 25sf51027@stu.hit.edu.cn; ykkttseng@hit.edu.cn; wenjiecoder@outlook.com).

SFT: delayed inputs, weighted sum

```
d0 <= x; d1 <= d0; ...
acc = 19*d0 + 28*d1 + ... + 37*d12;
y <= acc[15:0];
```

Input registers feed one weighted-sum expression.

RF: registered partial sums

```
a0 <= 19*x + a1;
a1 <= 28*x + a2; ... a12 <= 37*x;
y <= a0[15:0];
```

Each partial sum is stored in a register.

Fig. S1. Abbreviated assignments from the packaged FIR implementations. The SFT output sums products of delayed inputs; the RF implementation registers partial weighted sums. Full modules, including widths and reset logic, are provided in the runnable package. These candidates illustrate a structural difference and are not a causal ablation.

functional pass counts, paired interval, per-design scores, and measured selection curve. Adding `--simulate` runs the actual files `demo/sft.sv` and `demo/rf.sv` with Icarus Verilog against the shared reference. The gate uses 2 stimulus streams of 1024 inputs, repeats each stream after reset, and requires a common valid latency up to 40 cycles after a 8-sample warmup. Simulation traces and reset checks are retained in the output. This is a functional simulation check; it does not provide formal equivalence.

The package’s `README.txt` provides installation commands, expected values, file descriptions, and contact information. Candidate hashes link the example to its exact draw and archived physical record; the selected passing-boundary implementation reports are included for inspection. The analysis runs without a GPU or Vivado because it replays archived physical measurements. It does not retrain the policies or rerun implementation, and the model weights and complete physical trial directories are outside the package. These limits make the scope of the runnable demonstration explicit while allowing the numerical comparison and the functional example to be checked independently of the authors’ server environment.

TABLE SI
COMPLETED POST-PRIMARY RF VERSUS SFT COMPARISON. UTILITY IS THE ALL-DRAW BOUNDED ROUTED SETUP-PERIOD FREQUENCY, WITH FAILURES SCORED AS ZERO.

Policy	Passing draws	Pass (%)	Utility (MHz)
SFT	538/960	56.04	34.25
RF	449/960	46.77	81.95

TABLE SII
MEASURED BEST-OF- N UTILITY IN MHz, USING FIXED CONSECUTIVE BLOCKS WITHIN EACH GENERATION REPLICATE. SELECTION USES MEASURED PHYSICAL OUTCOMES.

N	SFT	RF
1	34.25	81.95
4	59.14	95.04
8	74.91	99.66
24	103.04	107.26

C. Completed comparison on fresh draws

The completed campaign evaluates 20 previously used designs with 48 new draws per policy and design. SFT has two sampling replicates from the same checkpoint; RF uses the existing training-seed-three and training-seed-four checkpoints, each generating 24 draws per design. All outputs use the shared functional gate and bounded routed setup-period search with Vivado 2026.1. Every functional or physical failure contributes zero, and each design receives equal weight. This post-primary comparison comes from a completed campaign that also evaluated a DPO arm. The present package isolates the RF versus SFT contrast requested for the supplement; it is not a matched reproduction of an external method or an independent test set, and its measurements are not pooled with the primary study.

RF reaches 81.95 MHz in failure-aware utility, compared with 34.25 MHz for SFT (Table SI). The paired difference is 47.71 MHz, with a 95% family-by-regime design-bootstrap interval of [30.00, 64.13] MHz from 100000 resamples. The RF replicate means are 99.11 and 64.79 MHz, compared with 35.38 and 33.12 MHz for the SFT sampling replicates. RF’s functional pass rate is lower: 46.77% versus 56.04%. The physical utility gain therefore coexists with a correctness tradeoff. The interval is conditional on these checkpoints and draws and does not capture uncertainty from retraining SFT or the entire pipeline. The endpoint measures bounded routed setup-period performance, not full timing signoff.

D. Selection at equal draw budgets

To separate policy quality from the opportunity to select among several outputs, Table SII reports the best measured result in fixed consecutive draw blocks within each replicate. Failures remain zero, and the same draw budgets are used for both policies. RF has higher utility at each reported budget, while the gap narrows as more candidates are available. At the largest budget, SFT reaches 103.04 MHz and RF reaches 107.26 MHz. This comparison grants the selector access to measured physical outcomes; it does not model an inexpensive

reranker or include training cost. Together, the example and the complete draw-level comparison support a shift toward physically faster outputs under the stated evaluation contract, while leaving the overall compute tradeoff and transfer to broader RTL workloads open.

SII. HISTORICAL TRAINING AND CONTEXTUAL COMPARISONS

A. Domain adaptation and re-anchoring

The studies collected here provide supporting evidence under their original protocols, separately from the sealed RF experiment. Table SIII compares the raw-base, supervised, and earlier optimized checkpoints, retaining the full declared draw budget for each. The earlier 30-design evaluation provides broader five-family coverage: its penalized endpoint rises from 73.0 to 176.7 MHz, with 27 of 30 designs improving. Feed-forward families show the most consistent gains, while IIR is mixed and median remains slow in absolute terms. Because its uncertainty analysis is historical and post-hoc, this study supports the breadth of the findings but is not pooled with the sealed result.

The historical re-anchored MLP policy (v9) raises utility from 73.18 to 192.35 MHz on its own matched SFT sample, with 29 of 30 designs improving. Its mean gain is 119.17 MHz. Oracle acceptance also increases (Table SIV). The earlier v8 and v9 evaluations use different training budgets and generated samples. Their difference cannot isolate the causal effect of re-anchoring, and training-only seed logs do not supply additional held-out physical replications.

B. Second-backbone study

Qwen-Coder repeats the complete domain-SFT and policy-optimization recipe on the earlier 22-design, 3-family universe. This is a separate earlier pipeline experiment, with its own SFT initialization and reward configuration. The main paper retains its overall and regime-level results because they provide positive evidence beyond a single backbone. They are not a second test of the canonical-feature RF package, and are not pooled with the sealed study.

TABLE SIII
EARLIER FIVE-FAMILY CAPABILITY LADDER. UTILITY IS FAILURE-PENALIZED TIMING-DERIVED FREQUENCY; CORRECTNESS DENOTES THE SAMPLED ORACLE VERDICT.

Checkpoint	Utility (MHz)	Oracle pass (%)
Raw base	1.8	2.8
SFT	73.0	88.7
Earlier policy (v8)	176.7	86.9

TABLE SIV
HISTORICAL MLP RE-ANCHORING EVALUATION ON ITS OWN COMPLETE DRAW SET. THIS IS SEPARATE FROM THE SEALED RF STUDY AND THE EARLIER V8 EVALUATION.

Checkpoint	Utility (MHz)	Oracle pass (%)
Matched SFT	73.18	90.97
Re-anchored policy (v9)	192.35	95.07

TABLE SV

DESCRIPTIVE SOURCE-FEATURE AVERAGES ON COMMON ORACLE-PASSING SUPPORT. THESE LEXICAL FEATURES ARE NOT PHYSICAL PIPELINE-DEPTH MEASUREMENTS.

Feature	SFT	RF	Correctness
Max. multiplications/statement	8.776	1.377	17.335
Structural accumulation score	2.887	14.627	0.248
Structural pipeline ratio	0.145	0.665	0.024

C. API prompting and HLS context

The archived API experiment uses the same accelerator tasks and oracle, with 8 samples per prompt arm. HLS uses engineer-written C++ and either no pragmas or engineer-selected pragmas. The reported RTL comparison selects the fastest measured optimized candidate. These budgets, inputs, and selection rules differ from the main policy-average comparison. Within the API experiment, a timing-focused prompt raises conditional F_{max} from 52.8 to 239.9 MHz while correctness changes from 27.1% to 22.9%; its penalized endpoint is 55.0 MHz. This contrast illustrates why conditional speed and per-draw utility need to be reported separately. It describes one API configuration under a smaller sampling budget and does not establish a general ranking of frontier models.

For selected implementation quality, the fastest measured optimized RTL candidate reaches 0.96× the geometric-mean throughput of engineer-pragma HLS over 19 comparable designs. It reaches 46.0× no-pragma HLS. Because HLS starts from engineer-written C++ and the RTL value is selected rather than policy-average, this comparison is context, not the causal baseline.

D. Earlier general-RTL evaluation

The earlier base-model comparison showed pass-at-one decreasing from 32.2% to 16.0% after domain SFT. That experiment uses different generation settings; its absolute values are not comparable with the matched RF extension and are not pooled. Together, the two experiments identify specialization costs at different training stages, without establishing performance on unseen industrial RTL.

III. STRUCTURAL, SAMPLING, AND STATISTICAL DIAGNOSTICS

A. Cross-arm structural features

The comparison uses 16 designs with oracle-passing support in all three arms. Within each design, features are averaged over passing candidate multiplicities; designs then receive equal weight. RF has smaller per-statement multiplier groups and larger structural accumulation and pipeline-ratio scores than SFT. Correctness-only shifts in the opposite direction. These source-derived features describe expressions and assignment patterns, including patterns available to the RF reward. They do not measure routed pipeline depth or establish an independent causal mechanism. The main paper’s implemented FIR pair connects this descriptive evidence to a physical example.

B. Finite-support sampling analysis

For a perfect-selector comparison, the empirical SFT distribution contains every measured candidate with its multiplicity plus a zero outcome for every failure. If $G_d(x)$ is its empirical cumulative distribution, the maximum of N independent draws has cumulative distribution $G_d(x)^N$. This exact curve assumes access to every draw’s measured timing-derived frequency before selection. It upper-bounds reranking on this finite sampled support, not on the unknown full SFT distribution. Draw count is not wall-clock cost: oracle calls, uncached correct-draw implementation attempts, and content-deduplicated implementations are reported separately. Training cost is not amortized into these inference-only counts.

To measure concentration, define the all-sample candidate mass $w_{p,d,m} = c_m/n_d$ and the dominant mass $\max_m w_{p,d,m}$. Normalizing over correct candidates gives $\bar{w}_{p,d,m} = c_m/\sum_j c_j$ and effective support

$$D_{p,d} = \left(\sum_m \bar{w}_{p,d,m}^2 \right)^{-1}. \quad (\text{S1})$$

Lower $D_{p,d}$ means that probability is concentrated on fewer correct strings, which does not by itself imply better hardware. We therefore examine it alongside measured candidate frequencies and ask whether SFT sampling already contains an implementation at least as fast as the optimized conditional mean. Among 16 designs with correct sampled strings under both policies, SFT’s measured support contains a candidate at least as fast as the RF conditional mean on 14. Mean inverse-Simpson effective correct-string support contracts from 3.78 to 2.67. These observations support a shift toward faster implementations on most designs with common support, although the finite samples and the sealed exceptions leave circuit novelty unresolved.

On the sealed distribution, one RF draw lies between the empirical SFT perfect-selector expectations for 17 draws (92.47 MHz) and 18 draws (93.66 MHz). At the upper integer crossing, the SFT workload is 18 generations and 36 oracle simulation streams per design, with 10.125 expected correct-draw implementation attempts without caching or 3.078 distinct implementations with content caching. These are empirical expected workloads, not measured runtime or speedup. They exclude policy-training cost and assume perfect post-route selection on observed support; they do not imply dominance over an unrestricted SFT search.

C. Family effects, endpoint decomposition, and seed dispersion

Table SVII shows how the post-primary effects vary by family. FIR, reverse FIR, polynomial, and recursive-filter point estimates are positive, whereas median is negative and is represented only by extrapolation designs. These rows are descriptive because the family cells are small and unequal. The primary direction is nevertheless insensitive to two post-primary checks: the unstratified paired percentile bootstrap gives [31.1, 85.4] MHz, and the mean effect across leave-one-family-out omissions ranges from 46.0 to 69.7 MHz. Both

TABLE SVI

INDEPENDENT REPAIRED-RF TRAINING-SEED EFFECTS ON THE SEALED ENDPOINT. SEED ROWS ARE SEPARATELY REPORTED POINT ESTIMATES; THE SEED-AVERAGED ROW USES THE FROZEN PAIRED, STRATIFIED 95% BOOTSTRAP INTERVALS.

Estimate	Interpolation	Extrapolation	Overall
RF seed 1	75.6	23.4	49.5
RF seed 2	80.1	51.4	65.8
Seed average [95% CI]	77.9 [48.4, 107.2]	37.4 [13.9, 60.9]	57.6 [37.7, 77.2]

TABLE SVII

POST-PRIMARY DESCRIPTIVE FAMILY BREAKDOWN OF THE SEALED FAILURE-PENALIZED ENDPOINT. I/E GIVES INTERPOLATION/EXTRAPOLATION DESIGN COUNTS. ROWS ARE DESCRIPTIVE POINT ESTIMATES; THE SMALL AND UNEQUAL FAMILY CELLS DO NOT SUPPORT STABLE FAMILY-LEVEL CONFIDENCE INTERVALS.

Family	Designs	I/E	SFT [MHz]	RF [MHz]	RF-SFT [MHz]
FIR	5	3/2	26.6	112.8	86.2
Reverse FIR	5	3/2	6.8	28.2	21.4
Polynomial	4	2/2	32.7	136.8	104.1
Recursive filter	4	2/2	98.6	149.9	51.3
Median filter	2	0/2	7.8	4.5	-3.4

TABLE SVIII

POST-PRIMARY DECOMPOSITION ON THE COMPLETE SEALED DRAW SET. CONDITIONAL FREQUENCY POOLS CORRECT DRAWS, INCLUDING ZERO-VALUED IMPLEMENTATION FAILURES.

Policy	q (%)	μ (MHz)	$q\mu$ (MHz)
SFT	56.25	62.89	35.38
RF	53.85	172.71	93.01
MLP	49.90	132.56	66.14
Correctness only	67.40	36.62	24.68

checks are post- primary sensitivities, not replacements for the frozen interval.

Table SVIII separates functional yield from the physical quality of oracle-passing draws. With equal budgets across designs, pooled $q\mu$ reproduces the primary mean exactly; this pooled μ is not an unweighted mean of per-design conditional frequencies. RF increases pooled conditional frequency from 62.89 to 172.71 MHz while correctness decreases. In contrast, the correctness-only policy has a conditional frequency of 36.62 MHz. The observed RF gain is thus a shift in the physical quality of passing draws, not an increase in their count. All 364 additional-seed candidate records and 4011 trial records pass the artifact audit and exact offline replay of the frozen search. The RF seed mean is 96.06 MHz, with a paired gain of 59.38 MHz [39.21, 79.16]; its descriptive between-seed sample standard deviation is 4.35 MHz. The correctness-only mean is 24.78 MHz, with difference -11.91 MHz [-15.62, -8.38] and seed standard deviation 0.49 MHz. Unstratified intervals and every leave-one-family-out mean retain these directions. This strengthens the checkpoint replication evidence, but is neither a universal policy ranking nor an independent full-pipeline repeat. These newer-tool setup-period endpoints are not pooled with the original timing-derived results.

SIV. PHYSICAL-VALIDATION DETAILS

A. Board harness and feasibility amendments

The board study uses paired SFT/RF candidates from the same design and applies the same multiplicity-aware selection rule to both policies. Fig. S2 shows this selection contract, the implemented test harness, and the trace-validation path. All selected DUTs, an echo canary, and the selector interface occupy one bitstream/HWH pair. Each complete run sweeps the same clock grid, recomputes the passing boundary from raw scores, and records the bitstream, HWH, selector-map, and selection-manifest hashes. This common harness allows the paired outputs to be measured under the same clock and capture conditions.

The first outcome-independent family hash selected a reverse-FIR design with no correct candidate in either paired arm, so that attempt stopped without a silicon outcome. Before any board measurement, eligibility was amended to require a correct candidate in both arms and the same outcome-independent hash was reapplied. The common sweep floor was also lowered after sealed Vivado rows showed that the original range would not bracket every selected DUT. No reward, timing, area, power, or resource outcome was used to choose the replacement. Nevertheless, these feasibility amendments make the resulting board subset descriptive rather than independently confirmatory.

B. Selected setup-closure pilot

A separate pilot evaluates 10 selected candidates under Vivado 2026.1, applying constraints before synthesis and bracketing a setup-passing period for each candidate. It therefore changes both the tool version and constraint handling relative to the primary flow. The selected candidates yield Spearman correlation 0.963 between original timing-derived and setup-passing frequencies, with median symmetric percentage error 3.50%. None of the 3 non-tied pairs reverses direction (Table SIX). However, the largest candidate error is 45.56%: the reverse-FIR RF candidate changes from 211.19

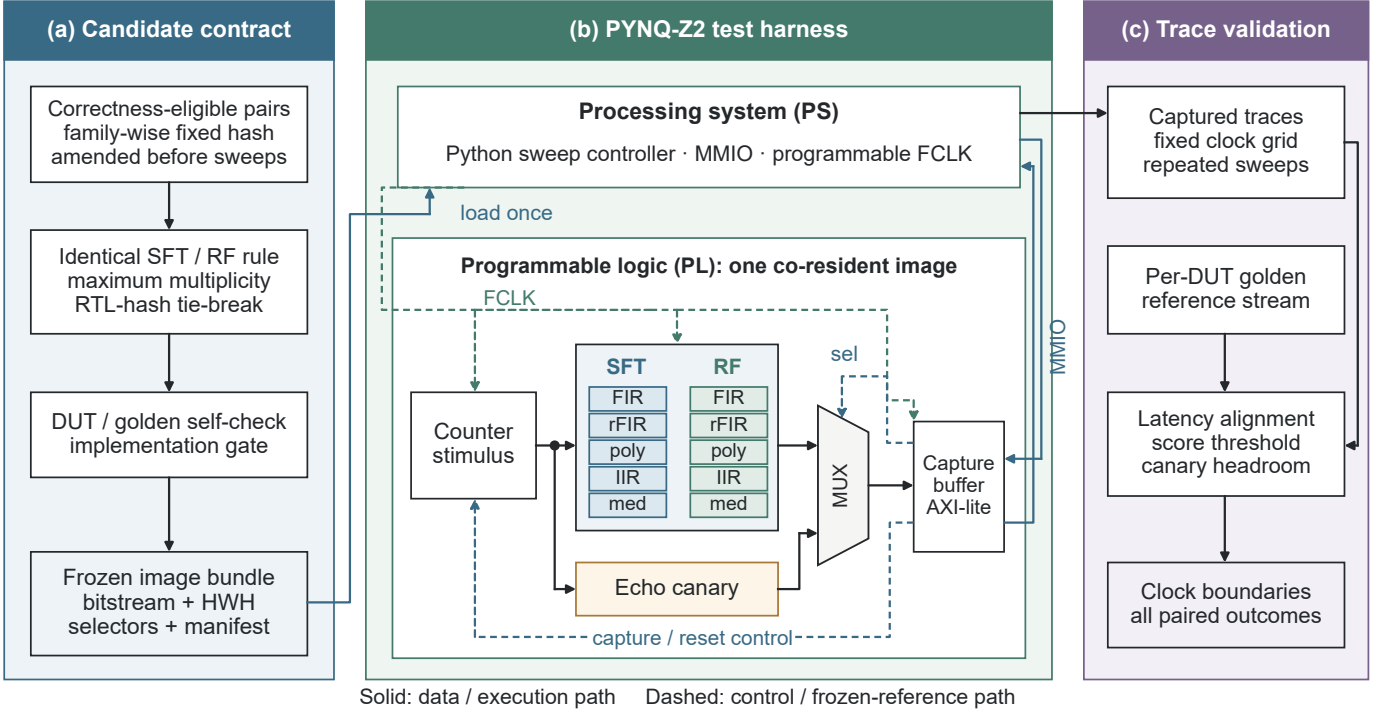


Fig. S2. Descriptive board-validation architecture. (a) The amended correctness-eligible selection uses the same multiplicity and hash rules for SFT and RF, with no frequency-based reselection. (b) The processing system configures the clock and accesses the capture buffer through memory-mapped AXI. Counter-driven SFT/RF DUTs and the echo canary coexist in one programmable-logic image; the selector chooses the captured output. Solid paths carry data or execution flow; dashed paths show clock and capture control. Logical buses are simplified for readability. (c) Captured traces are compared with per-DUT golden streams, and clock boundaries are accepted only with the specified canary headroom.

TABLE SIX

DESCRIPTIVE SETUP-CLOSURE SENSITIVITY (MHz). ORIGINAL: FROZEN WNS-DERIVED VALUE. SETUP: CONSERVATIVE PASSING ENDPOINT UNDER THE PILOT FLOW.

Design	SFT orig.	SFT setup	RF orig.	RF setup
fir42_v6_8b	20.49	20.65	179.40	185.19
firr20_v6	40.92	40.29	211.19	335.81
iir18_v7	184.84	177.89	184.84	177.89
med21	10.96	10.47	10.96	10.47
poly6_v13_8b	34.64	34.64	192.12	192.12

TABLE SX

CROSS-VERSION COVERAGE AND TIMING AGREEMENT. ERROR IS MEDIAN SYMMETRIC PERCENTAGE ERROR ON COMMON VALID CANDIDATES. UNRESOLVED CASES REMAIN INCONCLUSIVE.

Arm	Total	Valid	No path	Inconcl.	Error (%)
SFT	112	108	1	3	0.358
RF seed one	39	38	0	1	0.697
RF seed two	51	51	0	0	0.612

to 335.81 MHz. The median error therefore must not be read as a uniform accuracy guarantee. The check supports relative ordering on this subset, while absolute frequency remains sensitive to implementation conditions. Setup passing alone is not full timing sign-off.

C. Complete cross-version coverage diagnostic

The archived diagnostic preserves RTL, device, clock request, and constraint order while changing the Vivado version. It covers 202 original SFT/RF candidates. All 197 setup-path-present candidates have zero counts in the six named constraint-coverage checks. The confirmed missing-path case is sft_med21_g3; 4 other implementations remain inconclusive after the retained attempts. 2 SFT candidates also have multi-driver synthesis warnings; these are recorded without silently converting them into physical failures.

Median common-valid symmetric errors appear in Table SX; the maximum is 6.28%. The complete RF seed-two diagnostic mean is 100.67 MHz versus 101.14 MHz originally. SFT and RF seed-one means are unavailable because of the inconclusive cases, which are not assigned zero. Missing-path replays receive zero and all original draw denominators remain. These results describe timing sensitivity and coverage, not period-bracketed closure, synthesized functional equivalence, full timing sign-off, or a corrected primary confidence interval. The retained primary values are unchanged.

SV. REWARD DIAGNOSTICS AND VALIDATION HISTORY

A. Historical and repaired trajectories

Fig. S3 contrasts the older exploited reward with the repaired RF trajectory. In the historical run, early penalized physical performance rises from 76.4 to 230.5 MHz. Late in training, however, the proxy rises from 216.6 to 497.9 MHz

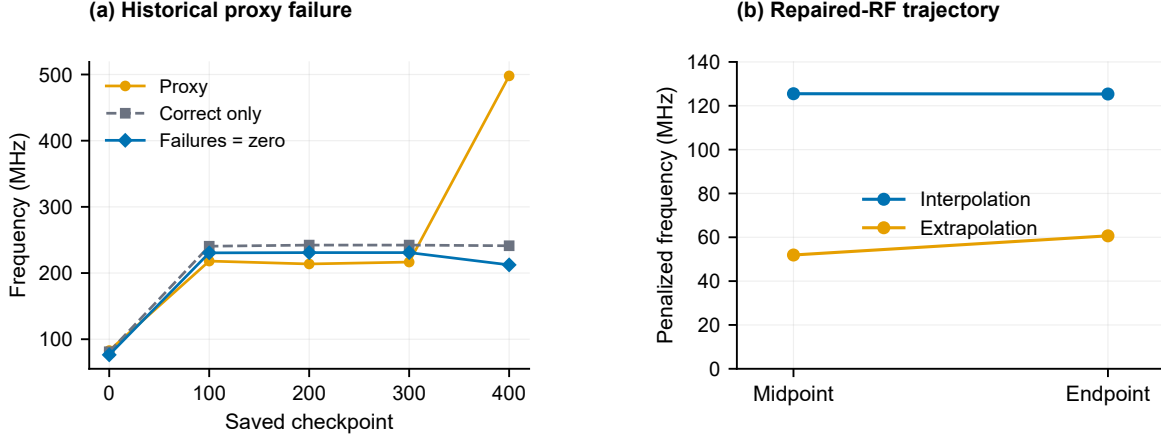


Fig. S3. Bounded reward-reliability evidence. Panel (a) shows a historical proxy rising while failure-penalized Vivado utility falls. Panel (b) shows the repaired RF midpoint-to-endpoint trajectory: interpolation changes from 125.5 to 125.4 MHz and extrapolation from 51.9 to 60.7 MHz.

TABLE SXI

ARCHIVED PREDICTOR RESCORING ON THE HISTORICAL OPTIMIZED DISTRIBUTION. BIAS IS IN MHz; CLAMP IS THE NUMBER OF SATURATED DESIGN CELLS; RHO IS CROSS-DESIGN SPEARMAN CORRELATION. ORIGINAL FITS WERE NOT RERUN FOR THIS REVISION.

Configuration	Bias	Clamp	ρ
mlp12_deployed	235.92	22	0.325
mlp12_refit	88.99	7	0.227
gbm12	27.13	0	0.326
rf12	-13.23	0	0.481
knn12	-11.12	0	0.543
ngram_ridge	-0.18	0	0.804
ngram_gbm	-18.81	0	0.455

while correctness falls from 95.3% to 88.0%. Penalized physical performance therefore falls from 230.9 to 212.3 MHz.

A lexical audit identifies one plausible channel in the older reward: a line-anchored assignment feature increases while its layout-canonical counterpart does not, and diagnostic rescoring removes 101.2% of the raw late reward jump. The canonical scores are clamp-saturated and there are 0 token-identical cross-checkpoint pairs, so this is bounded attribution rather than a randomized causal test. In the repaired run, interpolation remains essentially stable from midpoint to endpoint and extrapolation improves. The sealed efficacy study provides the evidence for the repaired reward. The historical trajectory explains the need for independent physical evaluation.

B. Archived predictor rescoring

The archive compares predictors on outputs produced by historical MLP-guided optimization. Table SXI retains the recorded fits and metrics; the models were not refitted in the original software environment for this revision. Several alternatives have smaller bias than the deployed MLP on this observed distribution, and the character n-gram ridge configuration has the highest cross-design rank correlation among these archived configurations. Because this comparison was made after the optimized outputs had been observed, it measures how well the predictors rescore a fixed distribution. It

does not show how policies trained against those alternatives would behave, isolate canonicalization in the sealed experiment, or establish resistance to future reward exploitation. The archive instead documents plausible predictor alternatives and motivates their prospective evaluation.

C. Executed reward path and contract history

The RF reward distinguishes 114 of 147 eligible RF groups, or 77.6%. Before sealed generation, all 801 distinct reward-eligible training candidates reproduced their logged canonical hash and features, compiled, and passed sampled original/canonical trace comparison; there were 0 rejections and 0 cross-design collisions. The trace check uses 256 vectors per frozen oracle seed and is evidence of the executed reward path, not formal equivalence. An earlier validation attempt used a stricter contract over every nonempty RF training completion. It stopped before sealed efficacy when that contract encountered a malformed, oracle-rejected completion. After diagnosing the scope mismatch, the present study froze a reward-eligible contract over exactly the candidates that could influence the RF reward, reused the already-trained checkpoints without retraining or selection, and opened a still-unseen sealed efficacy split only after the revised gate passed. The efficacy test is prospective with respect to that split, but it is not an independent validation of the post-diagnosis gate scope.

The revision's claim generator recomputes the historical re-anchored outcome from manifests, oracle summaries, and implementation rows; it revalidates structural and timing-coverage artifacts before rendering their claims. The predictor table is an explicitly archival diagnostic. Across both documents, generated ledgers retain source line ranges and hashes, while the manuscript verifier checks every included numerical claim and rejects stale outputs.